The Exceptional-Prime Desert Map

$\alpha_0 = 299 + \pi/10 \qquad$ rule: p exceptional $\iff r = p\,\|p\,\pi/10\| < 1$

Key constants

Quantity	Value
$\kappa = \varphi c/108$ (15 digits)	4.84330141945946
Bost bound $2\sqrt{13}$	7.21110255
Single-axis budget	0.600925
Classical set S_4	$\{2,\ 3,\ 19,\ 191\}$
$C_{\text{unwt}}(S_4) = C_{\text{unwt}}(S_{20})$	1.433676813 ($< 2\sqrt{13}$)
$C_{\text{wt}}(S_4)$	11.42214869
$C_{\text{wt}}(S_5)$ (at the bridge)	40.43789948
$C_{\text{wt}}(S_{20})$	40793.25685
κ-axis share at P_5	2.534298×10^9
Number of exceptional primes $< 10^{4000}$	20

H4 classical invariants $(I_k = \log p_k/(p_k - 1))$

The four real Weyl-axis shares of the 600-cell reading (other eight axes are zero).

Axis	Prime p	I_k	
1	2	0.693147	
2	3	0.549306	
3	19	0.163580	
4	191	0.0276435	
Sum $\sum I_k$		1.43368	(< 7.211

The bridge at P_5 (the one giant verifiable by hand)

$$P_5 = 3{,}993{,}746{,}143{,}633, \qquad q = 191\ (\in S_4), \quad m = 16, \quad k = 4302500806252,$$

$$\left|\, q\,\kappa^m - P_5 - k\pi \,\right| \;=\; 0.0382906 \;\pm\; 0.029 \;<\; 1, \qquad \text{error ratio} = 9.588 \times 10^{-15}.$$

For P_1–P_4 the relation is 0 exactly ($q = p,\ m = 0,\ k = 0$). For P_6–P_{20} it is *beyond tolerance* (propagated κ-uncertainty $10^{1.9} \ldots 10^{3537}$): marked **indeterminate**, never a pass.

Table 1 — The twenty exceptional primes

D_k = number of digits; w_k = desert width (run of consecutive non-exceptional integers ending just below p_k); $r_k = p_k\|p_k\pi/10\|$. Giants shown head...tail.

k	D_k	Prime p_k	Desert width w_k	r_k	Regime
1	1	2	0	0.7433629	Classical-Z
2	1	3	0	0.1725666	Classical-Z
3	2	19	15	0.5885052	Classical-Z
4	3	191	171	0.8441596	Classical-Z
5	13	3993746143633	3993746143441	0.1523629	**Bridge**
6	16	3224057731518397	3220063985374763	0.7749508	Desert
7	24	631474305334326148720631	631474302110268417202233	0.1441999	Desert

k	D_k	Prime p_k	Desert width w_k	r_k	Regime
8	35	`1053101266274469970227605594087344`	`10531012662113225396941729792152809`	0.3570549	Desert
9	76	`76365550274329238633...274704893143`	`76365550274329...64019701`	0.4366733	Desert
10	95	`87822897782612089843...462065410669`	`87822897782612...60517525`	0.8334696	Desert
11	111	`48983077336683228790...692784889819`	`48983077336683...19479149`	0.5405288	Desert
12	372	`19462609931099711093...516236611721`	`19462609931099...51721901`	0.5248112	Desert
13	859	`88321609940533775518...058013286951`	`88321609940533...76675229`	0.5359789	Desert
14	1025	`19681903024283226243...591371595031`	`19681903024283...58308079`	0.0147607	Desert
15	1592	`52943490324271006099...752677909607`	`52943490324271...06314575`	0.9803959	Desert
16	1863	`17494378896143817560...864658557771`	`17494378896143...80648163`	0.0504900	Desert
17	2272	`91511748796834962231...086060269167`	`91511748796834...01711395`	0.5457596	Desert
18	2389	`20716339208351233251...013122521601`	`20716339208351...62252433`	0.6931474	Desert
19	3428	`99469874438421417033...698029440779`	`99469874438421...06919177`	0.5105710	Desert
20	3548	`12737470687711639675...504644341493`	`12737470687711...14900713`	0.0276853	Desert

Table 2 — Weighted energy and bridge status

C_{wt} is cumulative; H4 = active symmetry axes; $\log_{10}\tau$ is the κ-tolerance (digits of κ needed to test the bridge).

k	C_{wt} (cum.)	H4	BDP status	error	$\log_{10}\tau$
1	1.386294	12	EXACT	0	—
2	3.034213	12	EXACT	0	—
3	6.142232	12	EXACT	0	—
4	11.42215	12	EXACT	0	—
5	40.43790	1	**VERIFIED**	0.0382906	−1.54
6	76.14732	1	BEYOND_TOLERANCE	—	1.6
7	130.9497	1	BEYOND_TOLERANCE	—	10.0
8	209.2893	1	BEYOND_TOLERANCE	—	20.4
9	384.0161	1	BEYOND_TOLERANCE	—	62.6
10	602.6319	1	BEYOND_TOLERANCE	—	81.8
11	857.5051	1	BEYOND_TOLERANCE	—	97.6
12	1712.430	1	BEYOND_TOLERANCE	—	358.7
13	3690.226	1	BEYOND_TOLERANCE	—	846.7
14	6048.751	1	BEYOND_TOLERANCE	—	1012.2
15	9713.830	1	BEYOND_TOLERANCE	—	1579.8
16	14001.80	1	BEYOND_TOLERANCE	—	1850.4
17	19233.19	1	BEYOND_TOLERANCE	—	2260.2
18	24732.49	1	BEYOND_TOLERANCE	—	2376.6
19	32625.75	1	BEYOND_TOLERANCE	—	3416.4
20	40793.26	1	BEYOND_TOLERANCE	—	3535.5

Honesty note. Desert widths and weighted energies are computed; the "bridge" is a numerical observation verifiable only at P_5 (and trivially P_1–P_4); the 12-curve H4 reading is an *interpretation*, not a theorem. The "error rate" (a measurement) and the "self-symmetry" (the 12 → 1 Coxeter reading) are distinct. No Hodge / BSD / mass-gap claim is made and no new mathematics is asserted. Certified data file `data/desert_map.csv` is byte-identical (sha256 `e0ea8b28...2291cbb`).